Huawei Access Node BAS-ACC-07 Site Basavanagudi POP Hardware/Environmental Fault Resolution SOP

Node: BAS-ACC-07 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_6054

Description

The Huawei Access Node BAS-ACC-07 at Basavanagudi POP is a critical component of the Telecom Network's access domain, responsible for terminating multiple optical fibers and converting them into electrical signals for transmission over the backbone network. The failure domain of hardware/environmental faults, specifically ASIC memory parity errors and line card crashes, are common issues that can occur due to the inherent complexity of the hardware and the environmental conditions in which it operates. The loss of power supply redundancy is also a critical failure mode that can occur due to various reasons such as power grid instability, overheating, or electrical surges. In this SOP, we will outline the diagnostic and resolution steps for addressing the Alarm Code ALARM_6054, which indicates a critical hardware/environmental fault in the access node.

Pre-requisite Checks

- Ensure remote hands are dispatched to Basavanagudi POP to assist with physical checks and troubleshooting

- Verify that the redundant link status is intact and functioning as expected

- Check the environmental conditions at the site, including temperature, humidity, and power supply redundancy

- Review the system logs to identify any recent errors or events that may be related to the failure

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to hardware failure or electrical surges	High
Line card crashes due to overheating or software bugs	Medium
Loss of power supply redundancy due to power grid instability or electrical surges	Low

Diagnostic Commands & Expected Outputs

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Command: display fault-management alarm, Expected Output: ALARM_6054: hardware/environmental fault detected, Description: This command will display the current alarm status and help identify the specific fault that triggered the ALARM_6054

Command: show tech-support, Expected Output: System logs, hardware information, and software versions, Description: This command will
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provide detailed information about the system, including logs, hardware configuration, and software versions, which can help identify the root cause of the failure

Command: display hardware, Expected Output: Hardware configuration, including line cards and power supplies, Description: This command will display the hardware configuration, including the status of line cards and power supplies, which can help identify potential issues
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Resolution Steps

Step 1: Step: Replace faulty line card, Command: replace line-card 1/1/1, Description: Replace the faulty line card with a new one to resolve the issue

Step 2: Step: Power cycle the system, Command: power-cycle system, Description: Power cycle the system to reset the configuration and resolve any software-related issues

Step 3: Step: Replace power supply unit, Command: replace power-supply 1/1/1, Description: Replace the faulty power supply unit with a new one to resolve the issue

Post-Check Verification

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Command: display fault-management alarm, Expected Output: No alarms detected, Description: Verify that the ALARM_6054 has been cleared and no new alarms are detected

Command: show tech-support, Expected Output: System logs, hardware information, and software versions, Description: Verify that the system logs, hardware information, and software versions are correct and up-to-date

Command: display hardware, Expected Output: Hardware configuration, including line cards and power supplies, Description: Verify that the hardware configuration, including line cards and power supplies, is correct and functioning as expected
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Roll-back Procedures

- Step: Restore original configuration, Command: restore configuration, Description: Restore the original configuration to its previous state in case the fix fails

- Step: Replace original hardware, Command: replace hardware 1/1/1, Description: Replace the replaced hardware with the original one to its previous state in case the fix fails